

IN6227 Data Mining - Assignment 1

Variant-2

IN6227-Assignment-1

Matric number: G2606362G Full name: NI QIXIN

Model name & version: deepseek/deepseek-flash LLM interface: Agent Harness such as opencode 1.18.34

Dataset: IN6227 A1 Variant 2 - ionosphere.csv (351 rows, 34 features, target class)

1. Data exploration and cleaning

Dataset. ionosphere.csv - 351 rows, 34 feature columns. The label is class with 2 classes, distributed g 225 / b 126 - a majority:minority ratio of 1.79:1, which is what the choice of metric in section 4 turns on. 34 numeric and 0 categorical columns were modelled; no column was held out.

Issues found. Missing values: none; duplicate rows: 1; outliers: kept - no row was filtered, so no fence was applied; class imbalance: 1.79:1.

Cleaning decisions.

Issue	Action	Why
No missing values	no imputation by hand; flag true (0 rows)	imputation stays inside each fold; nothing is missing.
Constant column attr_1	kept	no variance to leak or split on.
One duplicated row	kept	plausible coincidence; contradicted_rows 0.
Heavy-tailed values	keep all rows (no IQR)	b is the anomalous class, so filtering deletes signal.

After cleaning. 351 rows loaded; none removed; 351 rows modelled, 34 features.

2. Feature selection / engineering

None constructed: the 34 attributes are already normalised radar measurements, there are no missing values or categorical columns, and the constant and binary columns are neither identifier-like nor removed.

3. Model training

Candidate-model reasoning. Candidate_screen.json fitted all seven families on the same 280-row pool. SVM led (f1_macro 0.9189), then gradient boosting (0.9158) and random forest (0.9039); decision tree (0.8582), naive Bayes (0.8570), logistic regression (0.8434) and KNN (0.7677) fell below the one-standard-deviation floor and are eliminated on their measured defaults. Only svm+gradient_boosting and svm+random_forest differ in bias. SVM gives the kernel margin, suited to 34 standardised attributes and a locally separable b class (lift 1.89); gradient boosting takes the staircase slot over the lower-screened forest, since a second staircase adds no new failure mode.

	Model A	Model B
Estimator	Support vector machine (SVC via svm)	Gradient boosting (GradientBoostingClassifier via gradient_boosting)
Hyperparameters	C=10, gamma=0.1, kernel='rbf' - best of 18 combination(s) scored	learning_rate=0.05, max_depth=3, n_estimators=100 - best of 8 combination(s) scored
Tuned on	5-fold cross-validation over the 280 pool rows, scored on f1_macro	5-fold cross-validation over the 280 pool rows, scored on f1_macro
Validation score	0.9460 (f1_macro)	0.9242 (f1_macro)
Stopping criterion	libsvm runs to its default tolerance (tol=1e-3); no iteration cap was set.	fixed 100 boosting rounds from the grid; no early stopping.
Overfitting control	soft-margin C in {1,10,100} and gamma shrinkage bound the margin.	max_depth <= 3 and learning_rate shrinkage cap tree complexity.
Preprocessing applied	numeric: median imputation, then standardised; categorical: most-frequent imputation then one-hot (a level unseen in training encodes as all-zeros)	numeric: median imputation only, no scaling; categorical: most-frequent imputation then one-hot (a level unseen in training encodes as all-zeros)

Preprocessing, per model.

- svm: generally recommended, close to required with the rbf kernel - the margin is a distance, so an unscaled feature would set it; standardising stays inside the fold.
- gradient_boosting: not required - a split thresholds one feature, so monotone rescaling leaves every reachable split identical.

Fixed configuration (not tuned): seed 42; test fraction 20% of the frame; validation by 5-fold cross-validation

over the pool; tuning metric `f1_macro`; primary metric `f1_macro`.

4. Evaluation and comparison

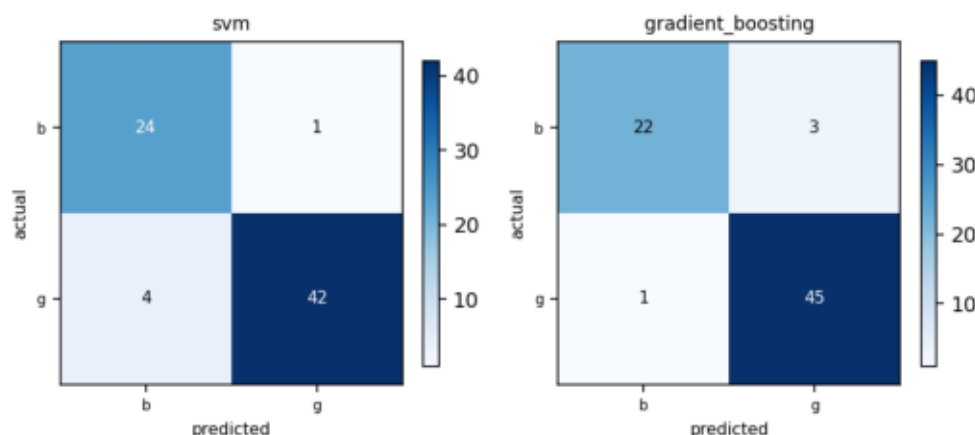
Split protocol: 280 pool rows, validated by 5-fold cross-validation, plus 71 test rows. The test set is stratified by class (20% of `ionosphere.csv`), seed 42, and was not touched again until the final scoring.

Validation strategy: stratified k-fold cross-validation, splitting at the row unit. Chosen because `structure.json` finds no entity key and no time column, so rows are independent and the unit is the row. Stratified 5-fold beats a holdout here because the grids hold 18 and 8 combinations: one split would overfit the choice, while 5-fold averages five estimates, validates every rare row once, and leaves the final fit on the whole pool. `test_size 0.2` holds out 71 rows for the final comparison. Validation class counts: b 20.2 / g 35.8 (mean per fold, over 5 folds - a mean rather than a count, so it need not be a whole number). Cost of the choice: 5x fitting; every row is validated against exactly once, so the score is the mean of 5 estimates rather than a single one, and the final model still trains on all 280 rows.

Validation used for: hyperparameter selection. Final model fitted on: the whole pool (280 rows) - with 5-fold cross-validation every row is validated against exactly once, so `final_fit_scope`: pool fits the same rows as train would, and the test set is the only estimate untouched by selection.

Metric	Model A	Model B
Accuracy	0.9296	0.9437
Precision (macro)	0.9169	0.9470
Recall (macro)	0.9365	0.9291
F1 (macro) (primary)	0.9247	0.9371
ROC-AUC / PR-AUC	0.9809 / 0.9876	0.9704 / 0.9758

Confusion matrices:



Primary metric and rationale. `f1_macro`. Tell good from bad radar returns; the split is uneven at 225:126 and both errors matter, so classes are weighted equally, and 126 rare rows make a macro average estimable. `f1_macro` is both headline and search objective, so selection and reporting share one ruler. Accuracy is rejected: the 64% majority dominates it. accuracy was not led with: dominated by the 64% majority g; an all-g predictor already reaches about 0.64.

5. Findings and discussion

Performance comparison. On the same 71 test rows, gradient boosting edged the SVM (`f1_macro` 0.9371 vs 0.9247; accuracy 0.9437 vs 0.9296) despite losing cross-validation (0.9242 vs 0.9460) - 71 rows is a coarse estimate. The SVM caught more b (24/25, recall 0.96) but made 4 g false positives; gradient boosting was more balanced (b recall 0.88, precision 0.96) with 4 total errors against 5. One or two rows separate them, inside the noise floor.

Limitations. 351 rows over 34 features makes the test estimate coarse: one flipped prediction moves macro-F1 by over a point and the test holds only 25 b rows. The single CSV was split internally, so no population shift is tested. The b class is real but locally separable, which can flatter the aggregate.

Repository

<https://github.com/NIQIXINNN/IN6227-assignment>